



Electronic Mail



Reti di Calcolatori Modulo 2

Introduction

- ▶ Electronic mail (e-mail)
 - ▶ Among most widely used Internet services
 - ▶ Use client-server paradigm
 - ▶ Allows users to send memos across the Internet
 - ▶ Small notes
 - ▶ Large memos
 - ▶ More file sent with e-mail than with FTP

Introduction

- ▶ Two major components
 - ▶ User interface
 - ▶ Mail transfer software
- ▶ Three major entities
 - ▶ User, that sends an e-mail
 - ▶ Client, a server from a user point of view that sends the e-mail out
 - ▶ Server, remote server receiving the e-mail and dispatching it to the destination
- ▶ Paradigm: transfer is a separate background activity

Mail delivery

- ▶ Different from other protocols (timeout and retransmission)
- ▶ The electronic mail works also if the remote machine is unreachable
 - ▶ The sender does not wait the response
 - ▶ The sender does not abort the transfer
- ▶ Delayed delivery (aka *spooling*)

Delayed Delivery: Spooling

- ▶ User sends a message
 - ▶ Message stored in the spool area (mail queue)
 - ▶ The system starts the transfer as a background activity
- ▶ Background activity becomes the client

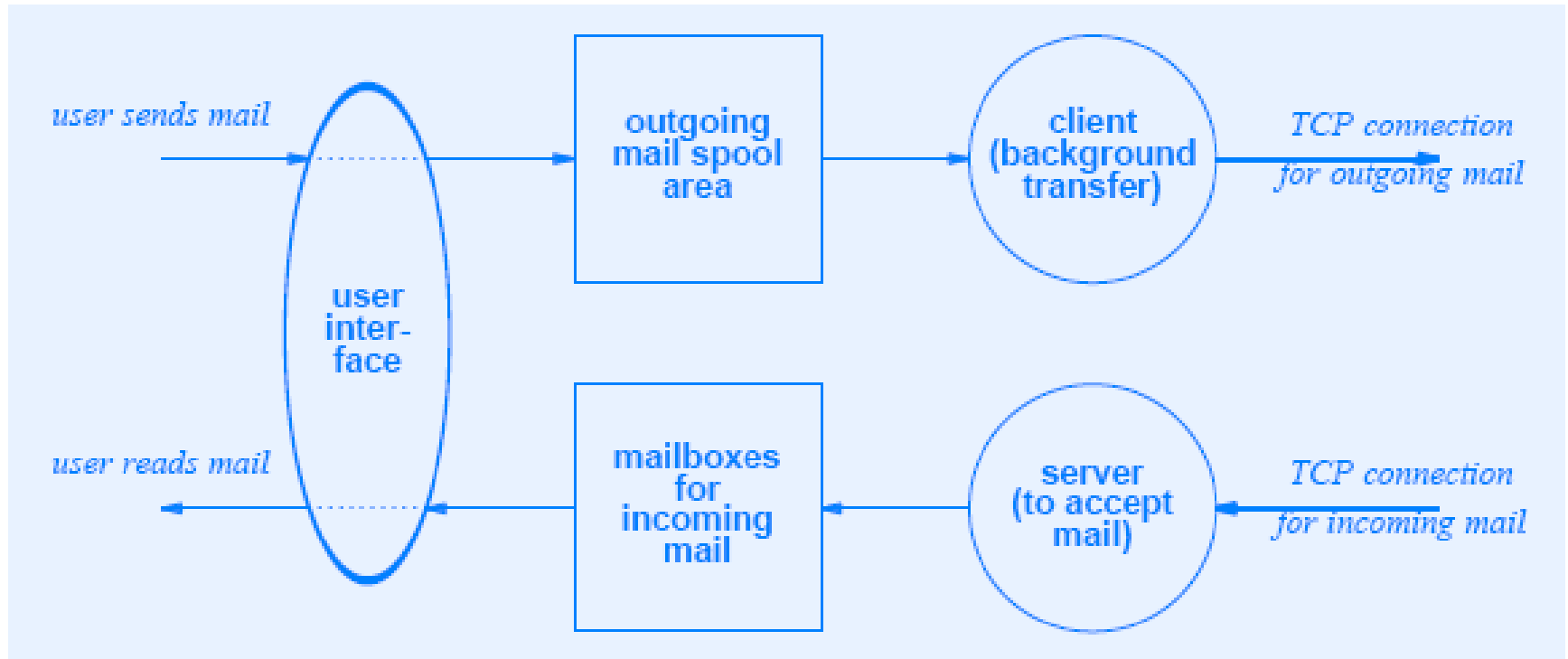
Delayed Delivery: Spooling

- ▶ Mail transfer process
 - ▶ DNS to map the destination name to IP address
 - ▶ TCP connection to the mail server (destination)
 - ▶ If ok, transfer the email to the server, which is stored in the remote spool area
 - ▶ Client removes the local copy
 - ▶ If connection is not possible, the transfer process stores the time and terminates

Delayed Delivery: Mail Retransmission

- ▶ The transfer process checks the spool area periodically (e.g., 30 minutes)
- ▶ For each mail not sent, a delivery process is tried
- ▶ A software error is returned after an extended time (e.g., 3 days)

Illustration Of Email System Components



Mailbox Names And Aliases

- ▶ Email destination identified by pair

(mailbox address, destination machine name)

- ▶ Aliases permitted (user enters alias that is expanded)
 - ▶ Names independent from other names on the machine (usually the same of login id but not necessarily)

Mailbox Names And Aliases

- ▶ DNS has a separate query type for mail destination
 - ▶ Decouple mail destination names from domain names
 - ▶ Telnet connection to example.com reaches a different machine from the one reached by an email to example.com

Alias Expansion and Mail Forwarding

- ▶ Powerful ideas
- ▶ Mail forwarding software includes a *mail alias expansion* mechanism
- ▶ Email arriving on a computer can be forwarded to one or more ultimate destinations

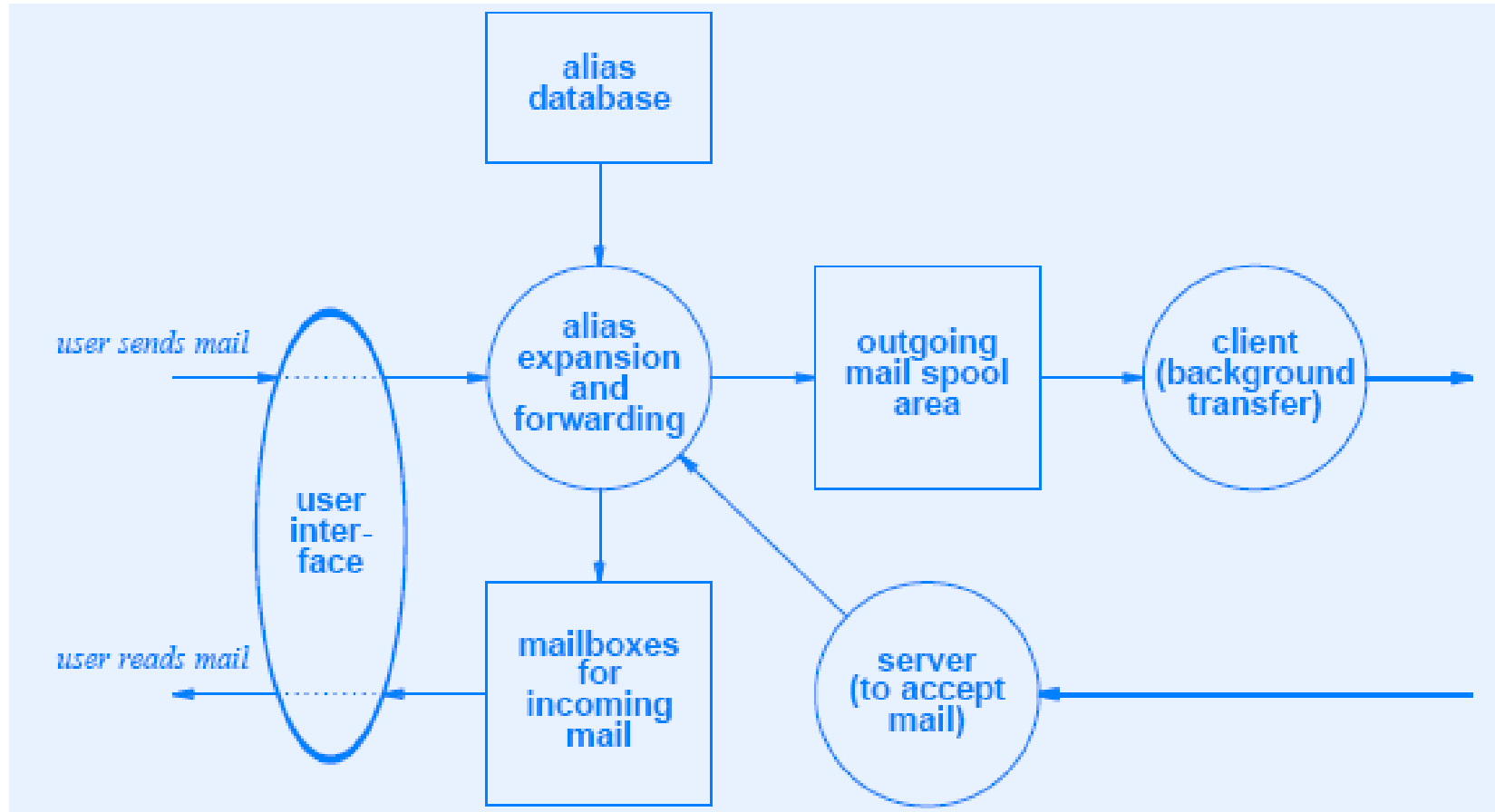
Mail Forwarding

- ▶ Mail forwarder maps identifiers to set of addresses
- ▶ The mail interface program expands the alias before passing it to the delivery system (user-client side)
- ▶ The same happen for the incoming message (server-side)

Forwarding

- ▶ Aliases increase functionality and convenience
- ▶ Many-to-one and one-to-many mapping
 - ▶ The first is different aliases for the same user
 - ▶ The second associates a group of users to a single alias (e.g., mailing list)
 - ▶ Used widely
 - ▶ Not all the recipient on a list need to be local
 - ▶ *Mail exploder* responsible for the mapping

Illustration Of Aliases And Forwarding



Aliases and forwarding: Problems

- ▶ Conflicting aliases
 - ▶ Site A maps mail x to mail y in Site B
 - ▶ Site B maps mail y to mail x in Site A
 - ▶ Result infinite loop
- ▶ Erroneous map
 - ▶ Users unable to read mail
 - ▶ If the mapped address exists, another user reads the mails
 - ▶ Otherwise the sender receives an error

TCP/IP Standards For Email

- ▶ Advantages
 - ▶ Universal delivery system
 - ▶ TCP/IP provides universal interconnection
 - ▶ Reliability
 - ▶ TCP/IP-based mail system more reliable
 - ▶ *Mail system that use end-to-end delivery can guarantee that each mail message remains in the sender's machine until it has been successfully copied to the recipient's machine*
 - ▶ TCP/IP guarantees end-to-end connectivity
- ▶ Alternative: *mail gateways*

Mail gateways

- ▶ Message transmitted through a chain of machines
- ▶ Sender's machine does not contact the destination one directly
- ▶ Sender sends the message to the first gateway that forwards to the second...

Mail gateways

- ▶ Advantage: interoperability
 - ▶ Provide connections among standard TCP/IP mail systems and other systems
 - ▶ Provide connections among TCP/IP networks and one that does not support Internet protocols
 - ▶ Companies outside Internet but able to communicate with email exchange

Mail gateways

- ▶ Disadvantages: unreliability and delay
 - ▶ The sender deletes the local copy of the message when delivered to the first gateway
 - ▶ Neither sender nor receiver have a copy of the message in transit
 - ▶ Failure or wrong forwarding results in loss of messages with the party being notified
 - ▶ Mail gateway can hold messages for a lot of time if the next hop is not available
- ▶ Sender and receiver must rely on machine that they do not control

TCP/IP Standards For Email

- ▶ Main goal provide interoperability
- ▶ Syntax for email addresses
- ▶ Two standards in the TCP/IP suite
 - ▶ Format of email messages
 - ▶ Protocols for email transfer and mailbox access

Email Address Syntax

- ▶ Mailbox identified by string

mailbox@computer

- ▶ String *computer* is domain name of computer on which a mailbox resides
- ▶ String *mailbox* is unique mailbox name on the destination computer

Format Of Email Message

- ▶ Message consists of
 - ▶ Header
 - ▶ Blank line
 - ▶ Body of message
- ▶ Headers have form
keyword: information
- ▶ Standard given in RFC 2822

Format Of Email Message: Header and Body

- ▶ Header can contain
 - ▶ *To:*
 - ▶ *From:*
 - ▶ *Reply-to:*
 - ▶ ...
- ▶ Body: text message
 - ▶ Avoid problems in binary representation

Protocol For Email Transfer

- ▶ Specifies interaction between transfer components
 - ▶ Transfer client
 - ▶ Transfer server
 - ▶ Exact format of messages to transfer mail
- ▶ Standard protocol is *Simple Mail Transfer Protocol (SMTP)*
 - ▶ Focuses on how the mail delivery system passes messages across the Internet
 - ▶ Does not consider how the system accepts or presents mails to the users, how mails are stored and retransmitted

SMTP

- ▶ Application-level protocol
 - ▶ Defined in RFC 821
 - ▶ Uses port 25
 - ▶ Uses TCP
- ▶ Commands and responses encoded in ASCII
 - ▶ Straightforward
 - ▶ Human readable
 - ▶ Rigid command format

Example of SMTP

```
S: 220 Beta.GOV Simple Mail Transfer Service Ready
C: HELO Alpha.EDU
S: 250 Beta.GOV

C: MAIL FROM:<Smith@Alpha.EDU>
S: 250 OK

C: RCPT TO:<Jones@Beta.GOV>
S: 250 OK

C: RCPT TO:<Green@Beta.GOV>
S: 550 No such user here

C: RCPT TO:<Brown@Beta.GOV>
S: 250 OK

C: DATA
S: 354 Start mail input; end with <CR><LF>.<CR><LF>
C: ...sends body of mail message...
C: ...continues for as many lines as message contains
C: <CR><LF>.<CR><LF>
S: 250 OK

C: QUIT
S: 221 Beta.GOV Service closing transmission channel
```

- ▶ Machine interprets 3-digit code and short commands
- ▶ Description for human comprehension
- ▶ Errors handled by the client
 - ▶ Usually reported as a mail

Example of SMTP

- ▶ Client can send a TURN command to turn connection
 - ▶ Client and server exchanged
 - ▶ The server sends any waiting mail message
- ▶ SMTP is more complex
 - ▶ E.g., if a user changes address and server knows its new address
 - ▶ It informs the client of the change
 - ▶ Forward directly the email to the new address or give the responsibility to the client

Protocols For Mailbox Access

- ▶ SMTP server must be always available
 - ▶ Otherwise it loses mails
- ▶ Used when user's mailbox resides on remote computer
 - ▶ Each user is assigned a mailbox on a computer with permanent Internet connection (SMTP server)
 - ▶ The user forms a temporary connection
 - ▶ The user runs the protocol to retrieve messages from the mailbox
- ▶ Especially helpful when user's local computer is not always on-line

Protocol For Mailbox Access

- ▶ Two protocols exist
 - ▶ *Post Office Protocol version 3 (POP3)*
 - ▶ *Internet Message Access Protocol version 4 (IMAP4)*
- ▶ Each provides same basic functionality
 - ▶ User authentication
 - ▶ Mailbox access commands

Post Office Protocol v3 (POP3)

- ▶ User invokes a POP3 client (e.g., thunderbird)
- ▶ POP3 client establishes a TCP/IP connection with a POP3 server
- ▶ User sends its *username/password*
- ▶ If accepted, user retrieves its mails
 - ▶ Retrieved mails are removed from the permanent mailbox
 - ▶ Message stored and transferred as text (RFC 822)

Internet Message Access Protocol v4 (IMAP4)

- ▶ Alternative to POP3
- ▶ As for POP3, *mailbox* is an abstraction located at the server
- ▶ User invokes an IMAP4 client (e.g., thunderbird)
- ▶ IMAP4 client contacts the server
- ▶ User can retrieve portion of the mails
 - ▶ Useful in case of low speed connections
- ▶ Retrieved mails are not removed from the permanent mailbox

SMTP and POP3

- ▶ The computer with the permanent mailbox needs of two server
 - ▶ Simple Mail Transfer Protocol (SMTP)
 - ▶ Defines how a mail system on one machine transfers mail to a remote server
 - ▶ POP3 or IMAP4 for messages in the permanent mailbox
 - ▶ Specifies how a users can extract messages from the mailbox

Multipurpose Internet Mail Extensions (MIME)

- ▶ Permits non-textual data to be sent in email
 - ▶ Graphics image, voice or video clip
- ▶ Sender
 - ▶ Encodes binary item into printable characters (ASCII)
 - ▶ Places in email message for transfer
- ▶ Receiver
 - ▶ Receives emails containing encoded item
 - ▶ Decodes message to extract original binary value

Multipurpose Internet Mail Extensions (MIME)

- ▶ MIME message includes
 - ▶ Type of the data
 - ▶ Encoding used to represent them
- ▶ MIME header
 - ▶ Resides in the RFC 822 mail header
 - ▶ Specify version, type, encoding

MIME Header

- ▶ Header in email message describes encoding used
- ▶ Example: jpeg using base64 representation

From: bill@acollege.edu

To: john@example.com

MIME-Version: 1.0

Content-Type: image/jpeg

Content-Transfer-Encoding: base64

...data for the image...

Seven Basic MIME Types

Content Type	Used When Data In the Message Is
text	Textual (e.g. a document).
image	A still photograph or computer-generated image
audio	A sound recording
video	A video recording that includes motion
application	Raw data for a program
multipart	Multiple messages that each have a separate content type and encoding
message	An entire e-mail message (e.g., a memo that has been forwarded) or an external reference to a message (e.g., an FTP server and file name)

MIME Types

- ▶ Composed of two identifiers:
 - ▶ Content type (e.g., image)
 - ▶ Subtype (e.g., gif)
- ▶ MIME permits a sender and receiver to define private content types
 - ▶ In addition to standard types

MIME Multipart Message

- ▶ It adds flexibility
- ▶ Four possible types
 - ▶ *Mixed*: multiple submessages with independent type and encoding (e.g., text and video)
 - ▶ *Alternative*: a single message includes multiple representation of the same data (e.g., message in plain text or html-formatted)
 - ▶ *Parallel*: a single message that includes subpart viewed together (e.g., video and music)
 - ▶ *Digest*: a single message that contains a set of messages

Example Of Mixed / Multipart Message

From: bill@acollege.edu

To: john@example.com

MIME-Version: 1.0

Content-Type: Multipart/Mixed; Boundary=StartOfNextPart

--StartOfNextPart

Content-Type: text/plain

Content-Transfer-Encoding: 7bit

John,

Here is the photo of our research lab I promised
to send you. You can see the equipment you donated.

Thanks again, Bill

--StartOfNextPart

Content-Type: image/jpeg

Content-Transfer-Encoding: base64

...data for the image...

Summary

- ▶ Email operates at application layer
- ▶ Conceptual separation between
 - ▶ User interface and message format
 - ▶ Mail transfer components

Summary

- ▶ Simple Mail Transfer Protocol (SMTP)
 - ▶ Standard for transfer
 - ▶ Uses ASCII encoding
- ▶ Post Office Protocol (POP3)
 - ▶ How to access the mails in the mailbox
- ▶ Multipurpose Internet Mail Extension (MIME)
 - ▶ Allows to send arbitrary data type using SMTP





Electronic Mail



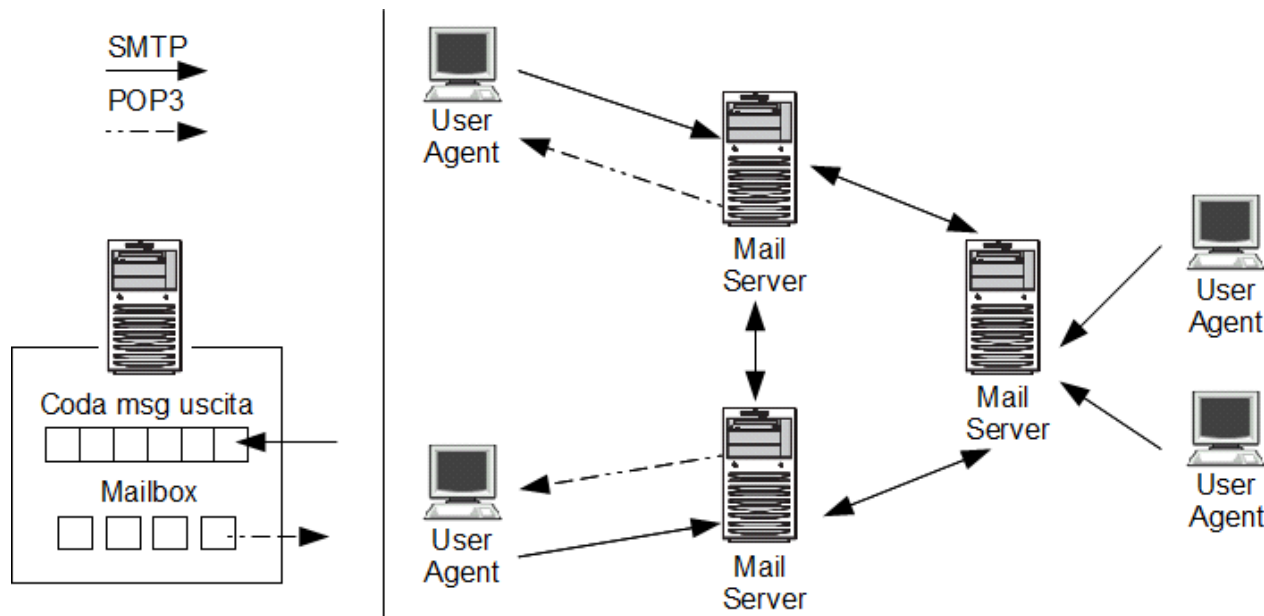
Approfondimenti

Posta elettronica

- ▶ La posta elettronica è una delle principali infrastrutture applicative su Internet e sulle reti IP aziendali
- ▶ Gli elementi principali del sistema di recapito della posta sono
 - ▶ User agent
 - ▶ Mail server
 - ▶ Mail box

Mail server

- ▶ I server contengono
 - ▶ Una coda messaggi composta dai messaggi di posta elettronica in uscita
 - ▶ Le mailbox contenenti i messaggi per ciascun utente



Protocollo SMTP

- ▶ SMTP (Simple Mail Transfer Protocol) consegna la posta ai server
 - ▶ Consegna dal client al server di posta locale
 - ▶ Consegna tra mail server
- ▶ SMTP è attivo tra agenti e server di posta per inviare messaggi di posta elettronica
 - ▶ Client: l'agente o il mail server che invia
 - ▶ Server: il mail server che riceve

Protocollo SMTP

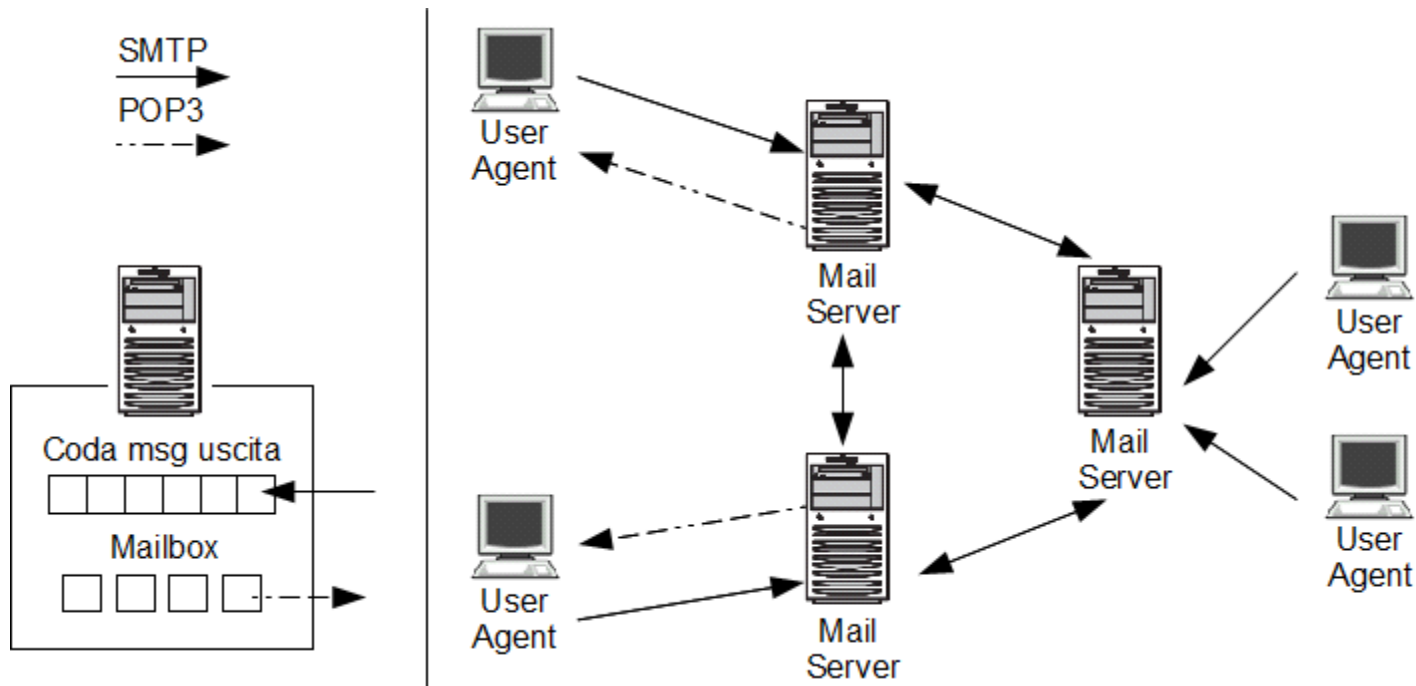
- ▶ SMTP usa un socket TCP sulla porta 25 per trasferire la posta elettronica in modo affidabile da client a server
- ▶ La posta elettronica viene memorizzata temporaneamente sul server locale e alla fine trasferita direttamente al server destinatario
 - ▶ La ritrasmissione intermedia è un caso non frequente

Protocollo SMTP

- ▶ Le tre fasi del protocollo sono
 - 1) Handshaking (detto anche HELO o saluto)
 - 2) Trasferimento di messaggi
 - 3) Chiusura
- ▶ L'interazione client/server segue un paradigma comando/risposta
 - ▶ I comandi sono in normale testo ASCII
 - ▶ Le risposte sono un codice di stato e una frase opzionale
 - ▶ Le righe di comando e risposta terminano con i caratteri CR-LF

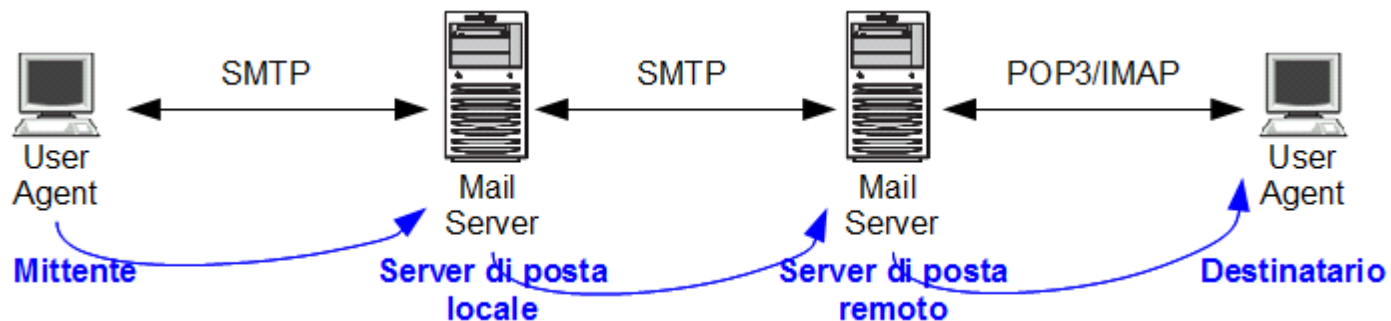
Protocollo POP

- Il POP (Post Office Protocol) consente l'accesso dell'utente alla posta elettronica consegnata alla sua mailbox



Consegna posta elettronica: il processo

- ▶ L'agente di posta dell'utente contatta via SMTP il suo mail server locale
- ▶ Il mail server locale contatta via SMTP il o i mail server di destinazione
- ▶ Il mail server di destinazione pone la posta nel mailbox dell'utente destinatario
- ▶ L'utente recupera la posta attraverso un protocollo di accesso alla posta come POP



Comandi SMTP Principali

HELO	Indica il nome del client
MAIL	Identifica il mittente
RCPT	Invia l'indirizzo di un destinatario
DATA	Invia i dati
RSET	Abortisce la trasmissione
VRFY	Chiede al server di verificare l'indirizzo di un destinatario senza spedire il messaggio
NOOP	Non fa nulla ma obbliga il peer a rispondere con OK
QUIT	Chiude la connessione
TURN	Inverte il senso della comunicazione

Esempio di una interazione SMTP

- ▶ Il client SMTP stabilisce la connessione TCP con il server pippo.edu alla porta 25

```
Server: 220 pippo.edu
Client: HELO pluto.it
S: 250 HELLO pluto.it, pleased to meet you
C: MAIL FROM:<alice@pluto.it>
S: 250 alice@pluto.it... Sender ok
C: RCPT TO:<bob@pippo.edu>
S: 250 bob@pippo.edu... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you know topolino?
C: And minnie?
C: .
S: 250 ok
C: QUIT
S: 221 pippo.edu closing connection
```

La riga con un solo punto è il delimitatore di messaggi.



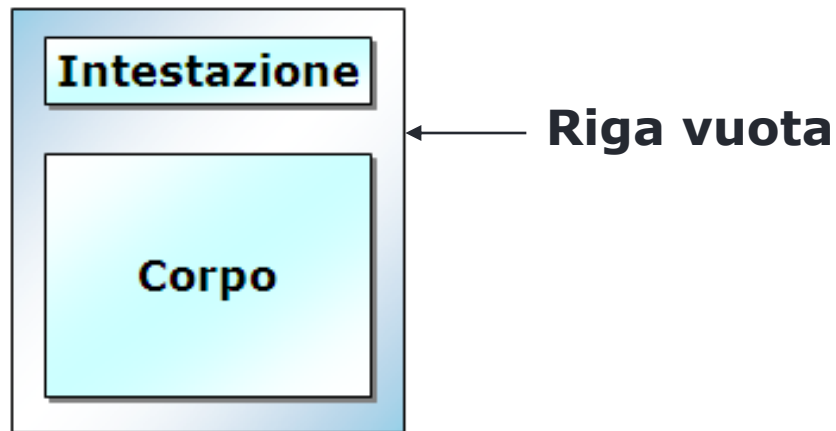
MIME



Approfondimenti

Formato dei messaggi di posta

- ▶ Righe d'intestazione, per esempio:
 - ▶ From:
 - ▶ To:
 - ▶ Subject:
- ▶ Corpo:
 - ▶ il “messaggio”, composto dai caratteri ASCII



Estensioni multimediali MIME

- ▶ SMTP richiede che tutti i dati siano caratteri ASCII a 7 bit
 - ▶ Tutti i dati non ASCII devono essere codificati come stringhe ASCII
- ▶ Ulteriori righe nell'intestazione del messaggio dichiarano il tipo di contenuto MIME.

Versione MIME

**Metodo usato per
decodificare i dati**

**Tipi di dati multimediale,
sottotipo,
dichiarazione parametri**

Dati codificati ASCII

```
From: alice@pluto.it
To: bob@pippo.edu
Subject: Picture of topolino
MIME-Version: 1.0
Content-Transfer-Encoding: base 64
Content-Type: image/jpeg

base64 encoded data .....
.....
.....base64 encoded data
```

Tipi di MIME

- ▶ Testo
 - ▶ Sottotipi: plain, html
- ▶ Immagini
 - ▶ Sottotipi: jpeg, gif
- ▶ Audio
 - ▶ Sottotipi: basic (8-bit μ -law encoded), 32kadpcm (32 kbps ADPCM)

Tipi MIME

- ▶ Video
 - ▶ Sottotipi: mpeg, quicktime
- ▶ Dipendenti dall'applicazione
 - ▶ Altri dati che devono essere elaborati dal lettore prima che siano “visualizzabili”
 - ▶ Sottotipi: msword,octet-stream

```
Content-Type: <type>/<subtype>[; <parameters>]
```

```
Content-Type: text/plain; charset=us-ascii
```

```
Content-Type: application/pdf; filename=foo.pdf
```

Contenuti multipli

- ▶ Nel caso di messaggi composti da “testo + dati multimediali” si introducono dei separatori

Versione MIME

```
From: alice@pluto.it
To: bob@pippo.edu
Subject: Picture of topolino
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary=4692718
```

**Separatore
intestazione/corpo**

```
--4692718
Content-Transfer-Encoding: quoted-printable
Content-Type: text/plain
```

```
Dear Bob,
Please find attached topolino's picture.
```

```
--4692718
Content-Transfer-Encoding: base 64
Content-Type: image/jpeg
```

```
base64 encoded data .....
.....base64 encoded data
--4692718
```

Contenuti multipli

- È possibile anche inviare vari tipi di contenuti multimediali

**Versione
MIME**

**Tipi
multipli
misti**

Parte 1

Parte 2

```
From: alice@pluto.it
To: bob@pippo.edu
Subject: Picture of topolino
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary=4692718
```

```
--4692718
Content-Transfer-Encoding: quoted-printable
Content-Type: text/plain
Dear Bob,
Please find attached topolino's picture.
```

```
--4692718
Content-Transfer-Encoding: base 64
Content-Type: image/jpeg
```

```
base64 encoded data .....
.....base64 encoded data
--4692718
```

**Separatore
di parti**

**Codifica e
tipo
specificati
per ogni
parte**

Note su SMTP

- ▶ Usa connessioni persistenti
- ▶ E' un protocollo "push"
 - ▶ Il server contatta il client
 - ▶ Il client spedisce i file (client->server)
- ▶ Con le estensioni MIME, in un singolo messaggio multiparte si possono inviare più oggetti

Protocolli di accesso alla posta

- ▶ SMTP consegna al server del destinatario
- ▶ Il protocollo di accesso alla posta recupera i messaggi dal server per conto di un utente
 - ▶ POP [RFC 1939]: autorizzazione e download
 - ▶ HTTP: Hotmail, Google Mail, ecc.
 - ▶ IMAP (Internet Mail Access Protocol) [RFC 1730]:
 - ▶ Più complesso, perché più funzioni
 - ▶ Manipolazione dei messaggi che restano memorizzati sul server
 - ▶ Permette di esaminare solo l'header o parte del messaggio

Protocollo POP3

► Fase di autorizzazione

► Comandi client:

► **User**

Dichiara il nome utente

► **Pass**

Password

► Risposte server:

► **+OK**

► **-ERR**

Fase di
autorizzazione

```
S:  +OK POP3 server ready
C:  user alice
S:  +OK
C:  pass paperino
S:  +OK

C:  list
S:  1 522
S:  2 881
S:  .
C:  retr 10
S:  <message 10 contents>
S:  .
C:  dele 10
C:  retr 22
S:  <message 22 contents>
S:  .
C:  dele 22
C:  quit
S:  +OK
```

Protocollo POP3

► Fase di transazione

- **list**: elenca numeri di messaggio
 - **retr**: recupera il messaggio per numero
 - **dele**: cancella
 - **quit**: chiude sessione
- Fase di transazione →

```
S: +OK POP3 server ready
C: user alice
S: +OK
C: pass paperino
S: +OK
C: list
S: 1 522
S: 2 881
S: .
C: retr 10
S: <message 10 contents>
S: .
C: dele 10
C: retr 22
S: <message 22 contents>
S: .
C: dele 22
C: quit
S: +OK
```

SMTP e HTTP a confronto

- ▶ Tipologia del protocollo
 - ▶ SMTP è un protocollo push
 - ▶ HTTP è un protocollo pull
- ▶ Persistenza
 - ▶ SMTP usa connessioni persistenti
 - ▶ HTTP può usarle o meno (LEZIONE HTTP)
- ▶ Contenuto del messaggio/oggetto
 - ▶ Entrambi hanno codici di stato e interazione comando-risposta ASCII
 - ▶ SMTP di default richiede che i messaggi siano ASCII a 7-bit: messaggio di più oggetti inviato con un messaggio multiparte
 - ▶ HTTP può trasferire tutto: ogni oggetto è incapsulato nelle intestazioni di risposta

Esercizio

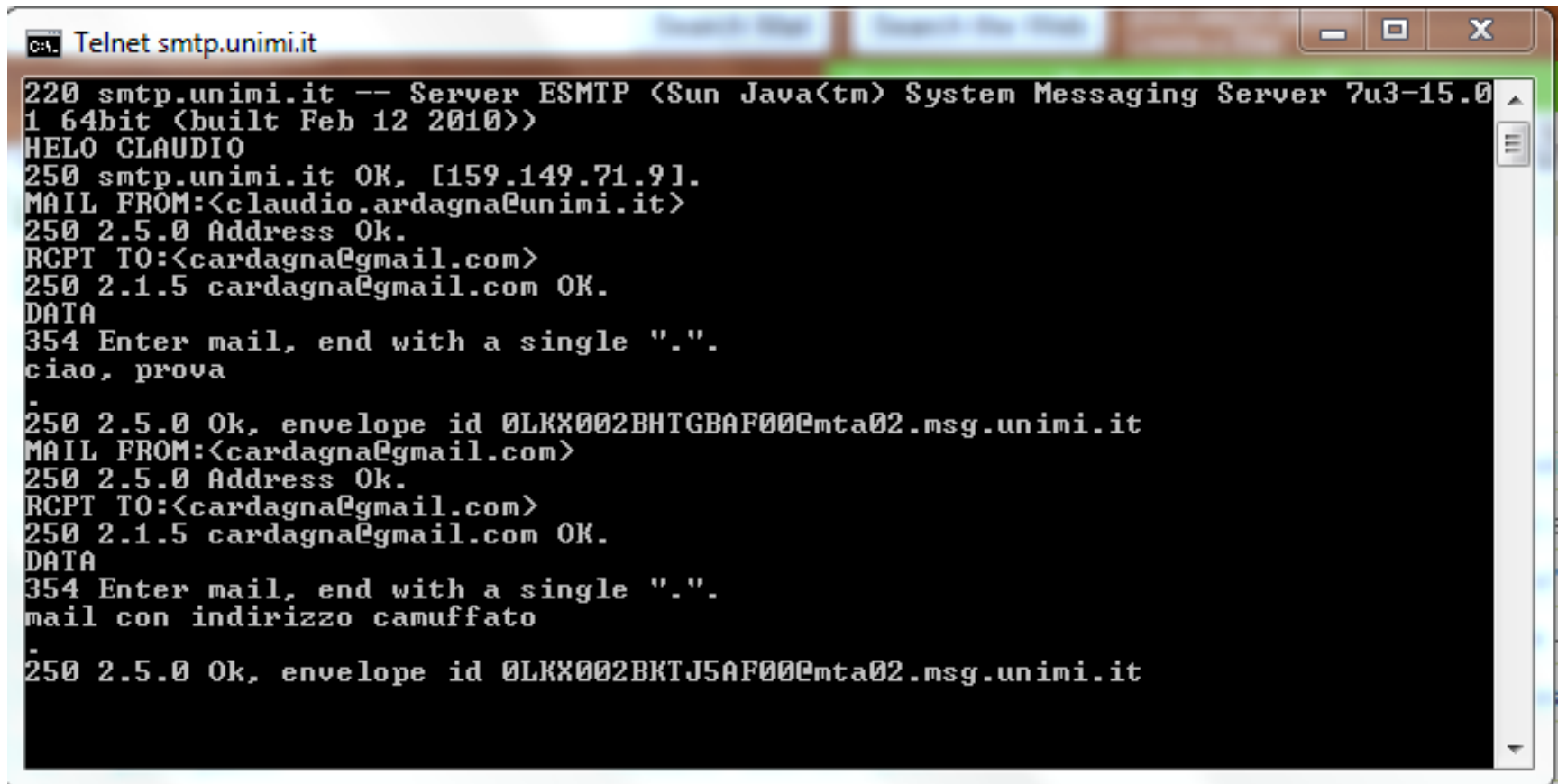
- ▶ Descrivere nel dettaglio il protocollo POP3 e le sue caratteristiche principali.
- ▶ Mostrare un esempio di comunicazione client-server nel quale un client richiede al server POP3 l'elenco dei suoi messaggi e legge il messaggio numero 5. Mostrare tutte le richieste e risposte.

Esercizio

- ▶ Mostrare tutte le richieste-risposte per l'invio di una mail tramite SMTP previa autenticazione.

Connessione a server SMTP

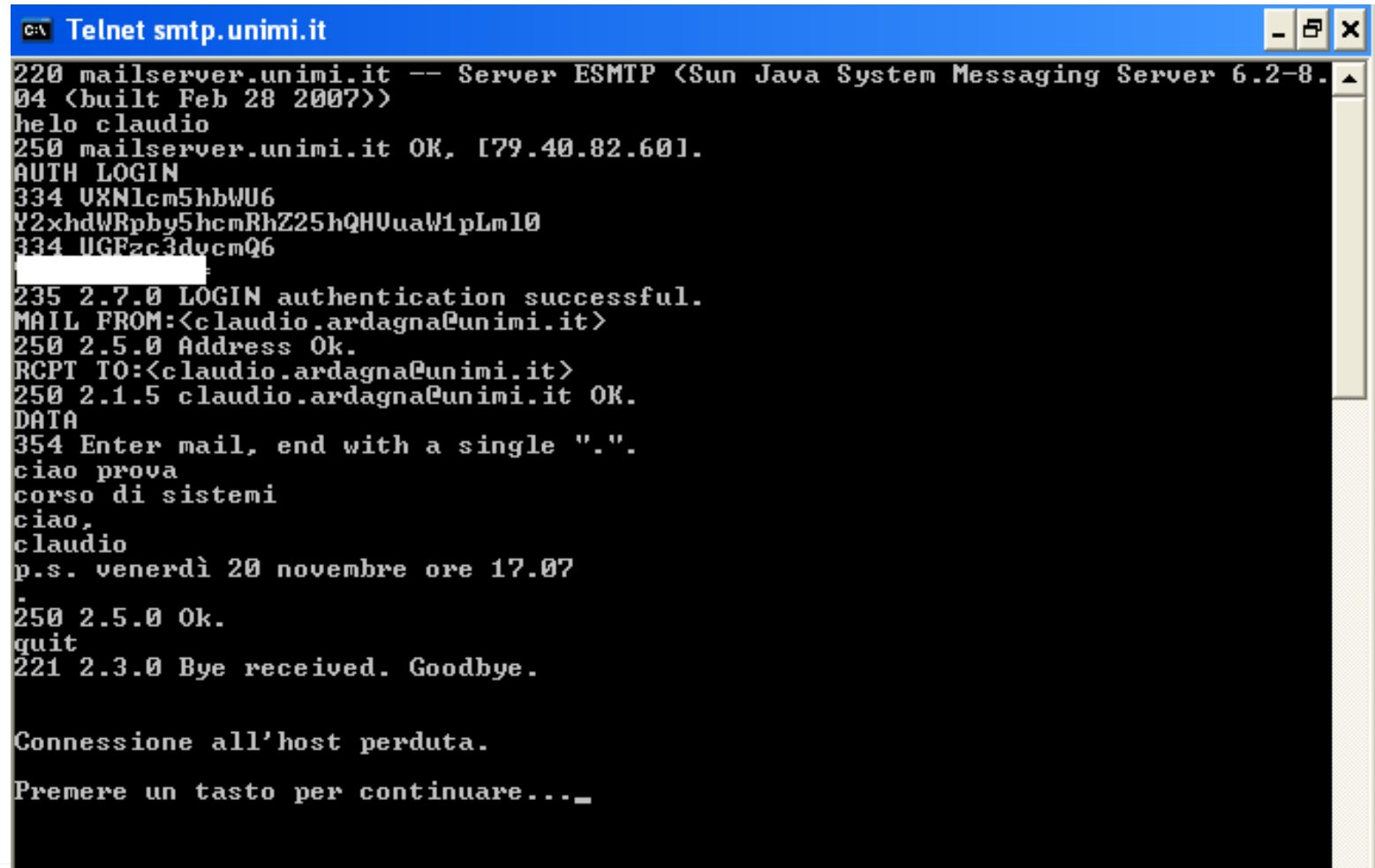
- ▶ telnet smtp.unimi.it 25
- ▶ Invio mail



```
Telnet smtp.unimi.it
220 smtp.unimi.it -- Server ESMTP (Sun Java(tm) System Messaging Server 7u3-15.0
1 64bit (built Feb 12 2010))
HELO CLAUDIO
250 smtp.unimi.it OK, [159.149.71.9].
MAIL FROM:<claudio.ardagna@unimi.it>
250 2.5.0 Address Ok.
RCPT TO:<cardagna@gmail.com>
250 2.1.5 cardagna@gmail.com OK.
DATA
354 Enter mail, end with a single ".".
ciao, prova
.
250 2.5.0 Ok, envelope id 0LKX002BHTGBAF00@mta02.msg.unimi.it
MAIL FROM:<cardagna@gmail.com>
250 2.5.0 Address Ok.
RCPT TO:<cardagna@gmail.com>
250 2.1.5 cardagna@gmail.com OK.
DATA
354 Enter mail, end with a single ".".
mail con indirizzo camuffato
.
250 2.5.0 Ok, envelope id 0LKX002BKTJ5AF00@mta02.msg.unimi.it
```

Connessione a server SMTP con pwd

- Username e password richiesti in base 64

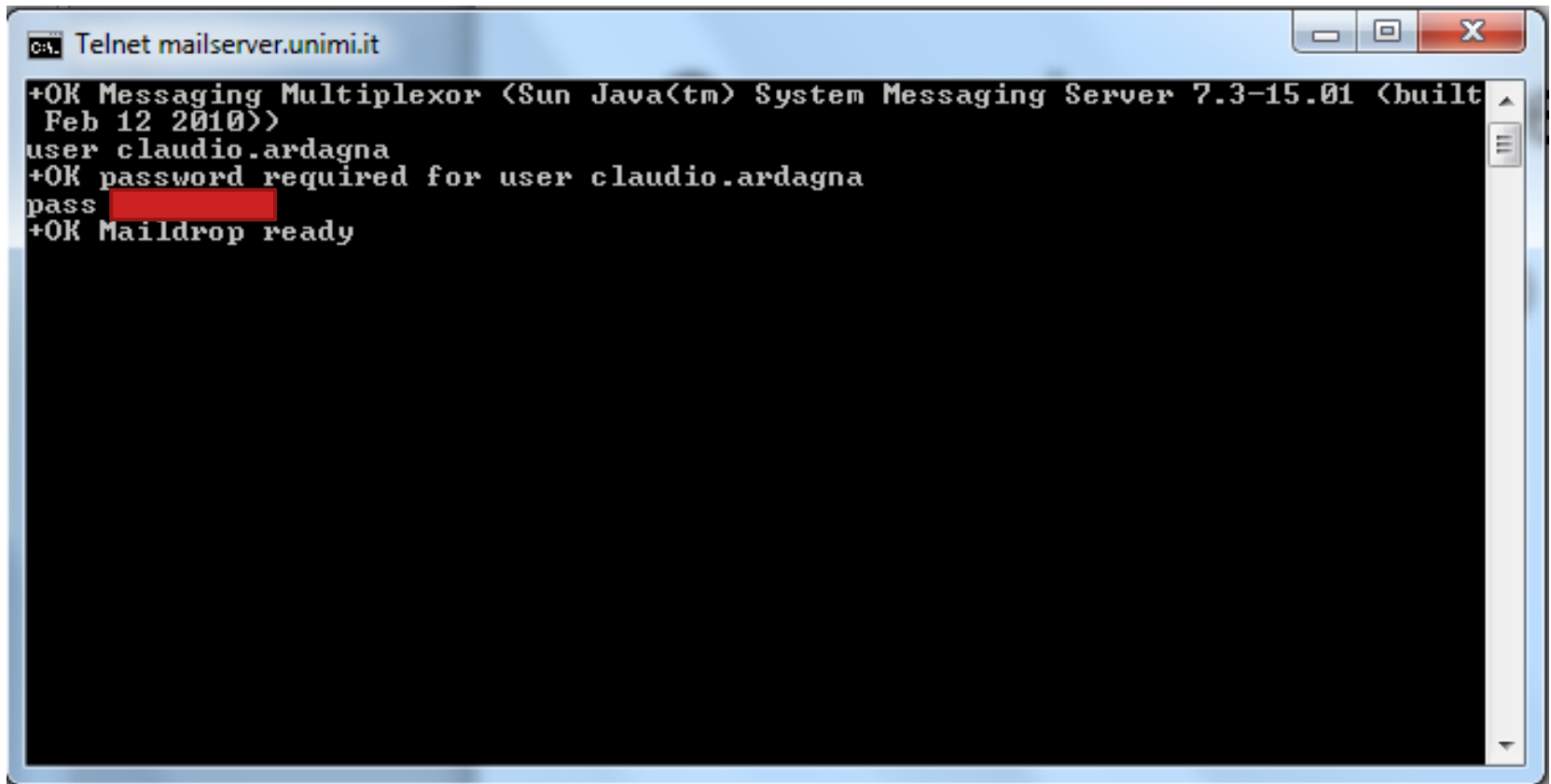


```
C:\ Telnet smtp.unimi.it
220 mailserver.unimi.it -- Server ESMTP (Sun Java System Messaging Server 6.2-8.
04 (built Feb 28 2007))
helo claudio
250 mailserver.unimi.it OK, [79.40.82.60].
AUTH LOGIN
334 VXNlcm5hbWU6
Y2xhdWRpby5hcmRhZ25hQHVuaW1pLml0
334 UGEzc3ducml0
[redacted]
235 2.7.0 LOGIN authentication successful.
MAIL FROM:<claudio.ardagna@unimi.it>
250 2.5.0 Address Ok.
RCPT TO:<claudio.ardagna@unimi.it>
250 2.1.5 claudio.ardagna@unimi.it OK.
DATA
354 Enter mail, end with a single ".".
ciao prova
corso di sistemi
ciao,
claudio
p.s. venerdì 20 novembre ore 17.07
.
250 2.5.0 Ok.
quit
221 2.3.0 Bye received. Goodbye.

Connessione all'host perduta.
Premere un tasto per continuare..._
```

Connessione a server POP3

- ▶ telnet mailserver.unimi.it 110



The screenshot shows a Windows command prompt window titled "Telnet mailserver.unimi.it". The window has standard Windows window controls (minimize, maximize, close) in the top right corner. The command prompt displays the following text:

```
+OK Messaging Multiplexor (Sun Java(tm) System Messaging Server 7.3-15.01 (built  
Feb 12 2010))  
user claudio.ardagna  
+OK password required for user claudio.ardagna  
pass [REDACTED]  
+OK Maildrop ready
```

The text is displayed in a monospaced font on a black background. The window title bar is light blue, and the command prompt itself has a black background with white text.

```

C:\ Telnet mailserver.unimi.it
1494 3054
1495 2308
1496 1952
1497 1875
1498 6602578
1499 1645
1500 2359
1501 3323
1502 3799
.
retr 1
+OK 2057 octets
Return-path: <Olga.Scotti@unimi.it>
Received: from unimix1.unimi.it ([172.24.4.81])
  by msg-1.unimi.net (Sun Java System Messaging Server 6.2-8.04 (built Feb 28
  2007)) with ESMTP id <0KFX007S5W4SXUD0@msg-1.unimi.net>; Tue,
  03 Mar 2009 17:50:52 +0100 (MET)
Received: from fara.unimi.net (unknown [172.24.3.21])
  by unimix1.unimi.it (Unimi Outbound) with ESMTP id 545EC558002; Tue,
  03 Mar 2009 17:50:52 +0100 (CET)
Received: from MonPetitPrince ([159.149.70.31])
  by ldap-s3.unimi.net (Sun Java System Messaging Server 6.2-8.04 (built Feb 28
  2007)) with ESMTP id <0KFX0095MW4RZWA0@ldap-s3.unimi.net>; Tue,
  03 Mar 2009 17:50:52 +0100 (CET)
Date: Tue, 03 Mar 2009 17:50:42 +0100
From: Olga Scotti <Olga.Scotti@unimi.it>
Subject: cartaceo FIRB
To: claudio.ardagna@unimi.it, marco.viviani@unimi.it, Antonia.Azzini@unimi.it,
  'Paolo Ceravolo' <Paolo.Ceravolo@unimi.it>
Cc: damiani@dti.unimi.it
Message-id: <46191A4010304D76B8BB3FC21C3E75E8@MonPetitPrince>
MIME-version: 1.0
X-MIMEOLE: Produced By Microsoft MimeOLE V6.00.2900.5579
X-Mailer: Microsoft Office Outlook 11
Content-type: text/plain; charset=iso-8859-1
Content-transfer-encoding: quoted-printable
Thread-index: AcmcIDCZmhh3B5+ilxqB6vefK1R6GA==
X-AuditID: 9f950a51-aa65ebb0000000bf9-fc-49ad5fec006f
X-Brightmail-Tracker: AAAAAGQ/idsNxfoj

Ciao,
Il cartaceo dei vostri modelli B e' stato pinzato insieme alla delibera =
di
ratifica del CdD e consegnato a Chiara Braghin che domani va a Milano a
consegnare tutto.
Per i due progetti in linea 1: la dichiarazione di impegno a farvi un
contratto triennale da giovane ricercatore viene redatta e spedita al =
MIUR
direttamente dal nostro Sportello Ricerca.
Ciao ciao, Olga

=BA=B0`=B0=BA=A4o=A4=BA=B0`=B0=BA=A4o=A4=BA=B0`=B0=BA=A4o=A4=BA=B0`=B0=BA=
=A4o=A4=BA=B0`=B0=BA=A4o=A4=BA=B0`=B0=BA
University of Milan, Department of Information Technologies
Via Bramante 65, 26013 Crema (CR), Italy=20
phone: +39-0373-898-063 or +39-02-503-30063=20
fax: +39-0373-898-010 web: http://www.dti.unimi.it/~oscotti/
=20

```